# lab 10

# DETERMINING THE $C_p / C_v$ RATIO OF GAS

## I. Aims / Purposes

- Investigating the mechanisms that after the thermodynamic equilibrium of air in a vase and calculating molecular heat capacity ratio of the gas.

## II. Apparatus, method and procedure

### a) Apparatus

- Large glass bottle with capacity 10L
- U-shape water manometer with milimeter-scale with accuracy 1mm
- Technical Brass Gas Ball valve K1, K2
- Rubber ball pump
- Support stand

### b) Methods

- We will use the Clement and Desormes method to determine the ratio of specific heats of air at constant pressure and constant volume, pressure thereby determining the $\frac{C_p}{C_v}$ ratio.

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c) Procedure.

- Step 1:
  - learn the opening and closing positions of valves K1, K2 by observation.
- Step 2:
  - Closed the valve $K_2$ and open the valve $K_1$
  - Pump air into flask A until the difference in the water column in the 2 branches of manometer M reaches about 250 - 300 mm $H_2O$, then closed the valve $K_1$
  - Wait 4-5mins for temperature and pressure reach a stable equilibrium
  - Measure many times with the same initial pressure $p_1$
  - Slowly open valve $K_2$ to reduce the amount of air in flask A so that the difference in water column in the 2 manometer branches M reaches the preselected value $H = L_1 + L_2$
  - Read the value of $L_1$ and $L_2$ and record in table 1.
- Step 3:
  - Quickly open valve $K_2$ to let the air in flask A escape
  - When the air presure in flask A equals the atmospheric pressure $H_0$ outside, we quickly closed valve $K_2$
  - For accurate measurement result, we need to observed and

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quickly close valve $K_2$ as soon as the water level in the 2 branches of the manometer M has just reached the same level, combined with the sound of air escaping from vase A has just stopped.

- Wait 4-5 mins for the temperature of the gas in flask A to equilibrium to room temperature.
- Then, the water head difference in the 2 branches M reaches a stable value $h = L_1' + L_2'$. Read the value $L_1'$ and $L_2'$ and record.

. Step 4:

- Repeat step 2 and step 3 5 times, corresponding to the select value of H.
- Record the corresponding measurment result of h in table 1

## III. Equations:

- The measured value and theoritical value of the ratio of air:

$$\varphi = \frac{H}{H-h}$$

$$\varphi_{theory} = \frac{i+2}{i}$$

- Calculating the relative deviations of $\varphi_{measured}$ and $\varphi_{theory}$:

$$\delta = \frac{\varphi_{theory} - \varphi_{measured}}{\varphi_{theory}}$$

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## IV. Experiment data

Table 1:

Accuracy of milimeter ruler on water column manometer

M: 1 (mm)

| | $H(mmH_2O)$ | $h(mmH_2O)$ | $\varphi = \frac{H}{H-h}$ | $\Delta\varphi$ |
|---|---|---|---|---|
| 1 | 256 | 74 | 1,41 | 0,002 |
| 2 | 256 | 74 | 1,41 | 0,002 |
| 3 | 256 | 74 | 1,41 | 0,002 |
| 4 | 250 | 72 | 1,40 | 0,008 |
| 5 | 252 | 74 | 1,42 | 0,012 |
| 6 | 268 | 78 | 1,41 | 0,002 |
| 7 | 280 | 82 | 1,41 | 0,002 |
| 8 | 274 | 80 | 1,41 | 0,002 |
| 9 | 296 | 86 | 1,41 | 0,002 |
| 10 | 272 | 78 | 1,40 | 0,008 |
| 11 | 252 | 72 | 1,40 | 0,008 |
| Avg | | | 1,408 | 0,005 |

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## V. Calculations:

- $\varphi_1 = \frac{H_1}{H_1 - h_1} = \frac{256}{256 - 74} = 1,41$

- $\varphi_2 = \frac{H_2}{H_2 - h_2} = 1,41$

- $\varphi_3 = \frac{H_3}{H_3 - h_3} = 1,41$

- $\varphi_4 = \frac{H_4}{H_4 - h_4} = 1,40$

- $\varphi_5 = \frac{H_5}{H_5 - h_5} = 1,42$

- $\varphi_6 = \frac{H_6}{H_6 - h_6} = 1,41$

- $\varphi_7 = \frac{H_7}{H_7 - h_7} = 1,41$

- $\varphi_8 = \frac{H_8}{H_8 - h_8} = 1,41$

- $\varphi_9 = \frac{H_9}{H_9 - h_9} = 1,41$

- $\varphi_{10} = \frac{H_{10}}{H_{10} - h_{10}} = 1,40$

- $\varphi_{11} = \frac{H_{11}}{H_{11} - h_{11}} = 1,40$

a) Average of $\varphi$

$$\bar{\varphi} = \frac{\varphi_1 + \varphi_2 + \varphi_3 + \varphi_4 + \varphi_5 + \varphi_6 + \varphi_7 + \varphi_8 + \varphi_9 + \varphi_{10} + \varphi_{11}}{11}$$

$$= 1,408$$

b) Calculating the absolute error:

$$\Delta\varphi_1 = |\bar{\varphi} - \varphi_1| = |1,408 - 1,41| = 0,002$$

- $\Delta\varphi_2 = |\bar{\varphi} - \varphi_2| = 0{,}002$
- $\Delta\varphi_3 = |\bar{\varphi} - \varphi_3| = 0{,}002$
- $\Delta\varphi_4 = |\bar{\varphi} - \varphi_4| = 0{,}008$
- $\Delta\varphi_5 = |\bar{\varphi} - \varphi_5| = 0{,}012$
- $\Delta\varphi_6 = |\bar{\varphi} - \varphi_6| = 0{,}002$
- $\Delta\varphi_7 = |\bar{\varphi} - \varphi_7| = 0{,}002$
- $\Delta\varphi_8 = |\bar{\varphi} - \varphi_8| = 0{,}002$
- $\Delta\varphi_9 = |\bar{\varphi} - \varphi_9| = 0{,}002$
- $\Delta\varphi_{10} = |\bar{\varphi} - \varphi_{10}| = 0{,}008$
- $\Delta\varphi_{11} = |\bar{\varphi} - \varphi_{11}| = 0{,}008$

Average absolute error:

$$\overline{\Delta\varphi} = \frac{\Delta\varphi_1 + \Delta\varphi_2 + \Delta\varphi_3 + \Delta\varphi_4 + \Delta\varphi_5 + \Delta\varphi_6 + \Delta\varphi_7 + \Delta\varphi_8 + \Delta\varphi_9 + \Delta\varphi_{10} + \Delta\varphi_{11}}{11} = 0{,}005$$

$H_{mid} = H_6 = 268$; $h_{mid} = h_6 = 78$; $\gamma_{mid} = \gamma_6 = 1{,}41$

$\Delta H_{mid} = \Delta L_1 + \Delta L_2 = 1 + 1 = 2$ (mm)

$\Delta h_{mid} = \Delta L_1' + \Delta L_2' = 1 + 1 = 2$ (mm)

$$\frac{\Delta\varphi}{\varphi} = \frac{H_{mid} \cdot \Delta h_{mid} + \Delta H_{mid} \cdot h_{mid}}{H_{mid}(H_{mid} - h_{mid})} = 0{,}014 = \alpha$$

$\Rightarrow \Delta\varphi_{sys} = \alpha \cdot \gamma_{mid} = 0{,}020$

$\Rightarrow \Delta\varphi = \Delta\varphi_{sys} + \overline{\Delta\varphi} = 0{,}020 + 0{,}005 = 0{,}025$

c) Results: $\varphi = \bar{\varphi} + \Delta\varphi = 1{,}408 \pm 0{,}025$

d) Calculating the theorical value of the $\varphi$ ratio of air according to the formula:

$i = 5$

$\Rightarrow \varphi_{theory} = \frac{i+2}{i} = \frac{5+2}{5} = 1{,}4$

$i = 6$

$\Rightarrow \varphi_{theory} = \frac{i+2}{i} = \frac{6+2}{6} = 1{,}333$

e) Comparing the measured value $\varphi_{measured}$ with the theorical value $\varphi_{theory}$ by calculating the relative deviations.

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$i=6$: $\delta = \dfrac{\varphi_{theory} - \varphi_{measured}}{\varphi_{theory}} = \dfrac{|1,333 - 1,408|}{1,333} = 0,056$

$i=5$: $\delta = \dfrac{\varphi_{theory} - \varphi_{measured}}{\varphi_{theory}} = \dfrac{|1,4 - 1,408|}{1,4} = 0,006$

VI. Questions:

1_ The state of gas mass is characterized by the following geometries: pressure p, volume V, temperature T

For 1 mole of gas:

$$pV = RT$$

where $R = 8,31$ J/mol.K in the gas constant

_ When heat is transfered to the air in the presence of mass m the air mass will heat up and its temperature donates an additional amount of $d_i$. The specific heat capacity of a substance when it is a quantity measured by the amount of heat that needs to be transfered to 1kg of the substance when its temperature is increased by 1degree, Therefore:

$$C_s = \frac{\delta . Q}{m dT}$$

For 1 molecular heat capacity of the gas:

$$C_M = M . C_s = \frac{\delta . Q}{dT}$$

$C_s$ (J.kg/K); $M$ (molarmass)(kg/mol)

= 0,005

12/5

7 cm

8

2 _ $C_V$ is the heat capacity during transformation where the volume of the system remains constant and is expressed in cell $\delta.Q_V$ divide by $ndT$

_ $C_p$ is the head capacity calculated during the transformation process where the pressure of the system remains constant and it expressed in cell $\delta.Q_V$ divided by $ndT$.

3 _ When the process compression or expansion of the gas happens very quickly, it does not have time to exchange heat with outside ($Q=0$), so it can be approximated as the process of compression or adiabatic expansion. When the process of compression or expansion of a gas takes place for a long time, it is considered isothermal compression or expansion.

_ Allow 4-5 mins in order to let temperature in flask A equilibrate to room temperature and water column difference reach stable value.

4 _ We have experimental conditions as: $h$ and $H \ll H_0$. We have $H_0$ of $Hg = 760$ mm $H_0$ of water is 10.3m. This demonstrates that measuring pressure with water is more sensitive than mercury. Therefore, it would be difficult to observe the

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change and measure $h$, $H$ if using mercury

- Experimental safety measurement: U-shaped tube with 2 open ends.

5. To ensure accurate measurement result, we must close $K_2$ quickly to make sure that the process is adiabatic (no transference of heat or mass to the surrounding). Then, the second reason why we need to closed $K_2$ as soon as the water column in the manometer equalize is because this is exactly when the internal pressure of gas equals the external pressure.

6. $$\frac{\Delta\varphi}{\varphi} = \frac{H.\Delta h + h.\Delta H}{H(H-h)}$$

we have:

$$\varphi = \frac{H}{H-h}$$

Take the natural logarithm on both sides:

$$\ln\varphi = \ln H - \ln(H-h)$$

Then take the differential:

$$\frac{\Delta\varphi}{\varphi} = \frac{\Delta H}{H} + \frac{\Delta H - \Delta h}{H-h} \Leftrightarrow \frac{\Delta\varphi}{\varphi} = \frac{+h\Delta H + H.\Delta h}{H(H-h)}$$

7. Since $O_2$ and $N_2$ are both diatomic gases, $i = 5$

$$C_V = \frac{i}{2}R = \frac{5}{2}R$$

$$C_p = C_V + R = \frac{5}{2}R + R = \frac{7}{2}R$$

$$\Rightarrow \varphi = \frac{C_p}{C_V} = \frac{7}{5} = 1,4$$

8. Use the formula:

$$\varphi = \frac{C_p}{C_V} = \frac{i+2}{i} = 1 + \frac{2}{i}$$

From the above formula, we can see that the theoretical value of the molecular heat capacity ratio of humid air will decrease compared to dry air because the humid air contains $H_2O$ which $i_{H_2O} = 6$ so $i$ of the system is not humid air will be larger than $i$ of dry air system